Chapter 2

Second Order Linear ODE

Learning Objectives

By the end of this chapter, students should be able to

1. distinguish between homogeneous and non-homogeneous second order linear ODEs;
2. solve homogeneous second order linear ODEs;
3. solve non-homogeneous second order linear ODEs using inverse D-operators;
4. solve engineering problems modelled by second order linear ODEs.

This chapter shows how we solve second order linear ODEs with constant coefficients. The method of using inverse D-operator is a simplified version of the Undetermined Coefficients (UC) method.

We start with a discussion of solving homogeneous second order linear ODEs in Section 2.1. Next, we introduce the inverse D-operator to solve non-homogeneous ODEs in Section 2.2. Applications of second order linear ODEs in engineering are demonstrated in Section 2.3 before we end with exercises in Section 2.4.

2.1 Homogeneous Second Order Linear ODE

A second order linear ODE is an ODE of the form $P_2 \frac{d^2y}{dx^2} + P_1 \frac{dy}{dx} + P_0\, y = Q$, where P_2, P_1, P_0, y, and Q are functions of x, with $P_2 \neq 0$. In this chapter, we consider second order linear ODEs with constant coefficients, that is, P_2, P_1, P_0 are all constants.

$$a \frac{d^2y}{dx^2} + b \frac{dy}{dx} + c\, y = Q$$

A second order linear ODE above is homogeneous if $Q = 0$, and non-homogeneous otherwise.

Definition 2.1.1 (Homogeneous Second Order Linear ODE and Characteristic Equation)

The ***homogeneous second order linear ODE with constant coefficients*** is of the form

$$a \frac{d^2y}{dx^2} + b \frac{dy}{dx} + c\, y = 0$$

and its ***characteristic equation*** is the quadratic equation

$$am^2 + bm + c = 0$$

(a, b, c are constants with $a \neq 0$).

Remark. Definition 2.1.1 can be extended to higher order linear ODE with constant coefficients. In some texts, the characteristic equation is also called the auxiliary equation.

The solutions of a characteristic equation are called characteristic roots. Recall that there are three types of solution for quadratic equations. The type of characteristic roots will determine the general solution of the homogeneous linear ODE.

Procedure in Solving Homogeneous Second Order Linear ODE

Step 1. Write the characteristic equation and solve for the characteristics roots.

Step 2. Let α and β be the characteristic roots. The general solution of the ODE is one of the cases below.

Type of characteristic roots	General Solution
Real with $\alpha \neq \beta$	$y = C_1 e^{\alpha x} + C_2 e^{\beta x}$
Real with $\alpha = \beta$	$y = (C_1 x + C_2)e^{\alpha x}$
Complex roots $p \pm qj$	$y = e^{px}(C_1 \sin qx + C_2 \cos qx)$

C_1 and C_2 are arbitrary constants.

Example 2.1.2
Solve $y'' - 4y = 0$.

Solution

Step 1. Characteristic Equation: $m^2 - 4 = 0 \Rightarrow m = 2$ or $m = -2$.
Step 2. The general solution is $y = C_1 e^{2x} + C_2 e^{-2x}$.

Example 2.1.3
Solve $y'' + 4y' + 4y = 0$.

Solution

Step 1. Characteristic Equation: $m^2 + 4m + 4 = 0 \Rightarrow m = -2$.
Step 2. The general solution is $y = (C_1 x + C_2)e^{-2x}$.

Example 2.1.4

Solve $\dfrac{d^2y}{dx^2} + 4y = 0$.

Solution

Step 1. Characteristic Equation: $m^2 + 4 = 0 \Rightarrow m = \pm 2j$.

Step 2. The general solution is $y = e^{0x}(C_1 \sin 2x + C_2 \cos 2x)$
or $y = C_1 \sin 2x + C_2 \cos 2x$.

Example 2.1.5

Solve the IVP:

$$\begin{cases} \dfrac{d^2y}{dx^2} + 4y = 0 \\ y(0) = 1 \\ y'(0) = 1 \end{cases}$$

Solution

From previous example, the general solution of $\dfrac{d^2y}{dx^2} + 4y = 0$ is

$y = C_1 \sin 2x + C_2 \cos 2x$.

$y(0) = 1 \Rightarrow 1 = C_1 \sin 0 + C_2 \cos 0 \Rightarrow C_2 = 1$

$y' = C_1 \dfrac{d}{dx}(\sin 2x) + C_2 \dfrac{d}{dx}(\cos 2x) = 2C_1 \cos 2x - 2C_2 \sin 2x$

$y'(0) = 1 \Rightarrow 1 = 2C_1 \cos 0 - 2C_2 \sin 0 \Rightarrow C_1 = 0.5$

The particular solution is $y = 0.5 \sin 2x + \cos 2x$.

2.2 Non-homogeneous Second Order Linear ODE

Definition 2.2.1 (Non-homogeneous Second Order Linear ODE)

The ***non-homogeneous second order linear ODE with constant coefficients*** is of the form

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=Q$$

where Q is a non-zero function of x, and a, b, c are constants with $a\neq 0$.

Observe that a non-homogeneous linear ODE can be written as

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=0+\mathrm{Q}.$$

If y_c is the solution of the corresponding homogeneous linear ODE

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=0$$

and y_p is a solution of

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=Q,$$

then the general solution of a non-homogeneous linear ODE is of the form

$$y=y_c+y_p$$

where y_c is the ***complementary function***, and y_p is the ***particular integral***.

To find y_c, we solve the corresponding homogeneous ODE as discussed in the previous section. To find y_p, we use inverse D-operators. An operator is a rule that assigns an input function to an output function. A D-operator takes differentiable functions as its inputs, and assigns the derivative of the input as the outputs, e.g. $D(\sin x)=\cos x$.

We use the abbreviated form D^2y to denote the second derivative of y. For instance, $D^2(\sin x)=D\big(D(\sin x)\big)=D(\cos x)=-\sin x$.

For our purpose, the linear ODE

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = Q$$

is written in the D-operator form as

$$(aD^2 + bD + c)y = Q,$$

and y_p is found as follows.

$$y_p = \frac{1}{aD^2 + bD + c}(Q)$$

The expression $\dfrac{1}{aD^2 + bD + c}$ above is an example of an ***inverse D-operator***, which takes the function Q as its input. For convenience, we use the notation $\dfrac{1}{F(D)}$ to denote an inverse D-operator.

We can determine the particular integral y_p if we know how the inverse D-operator acts on various types of input function Q:

1. $Q = k$, a constant function
2. $Q = e^{kx}$, an exponential function
3. $Q = e^{kx}\,V(x)$, a product of e^{kx} and constant/sine/cosine
4. $Q = \sin kx$ or $Q = \cos kx$ sine or cosine function

We will discuss all the cases above one by one.

Inverse D-operator of a Constant Function

$$\frac{1}{F(D)}(k) = \frac{1}{F(0)}(k)$$

In the formula above, $F(0)$ is the value obtained by replacing D in $F(D)$ with 0, provided that $F(0) \neq 0$. If $F(0) = 0$, we modify the method (see Example 2.2.3).

Example 2.2.2

Solve $\dfrac{d^2y}{dx^2} - 4y = 8$.

Solution

Find y_c: Consider $\dfrac{d^2y}{dx^2} - 4y = 0$. From Example 2.1.2, we have

$$y_c = C_1e^{2x} + C_2e^{-2x}$$

Find y_p: Consider $\dfrac{d^2y}{dx^2} - 4y = 8$.

$$(D^2 - 4)y = 8 \Rightarrow y_p = \frac{1}{D^2 - 4}(8)$$

$$\Rightarrow y_p = \frac{1}{0^2 - 4}(8) = \frac{8}{-4} = -2$$

Thus, the general solution is

$$y = y_c + y_p = C_1e^{2x} + C_2e^{-2x} - 2.$$

By the definition of the D-operator, it is obvious that $\frac{1}{D}\big(f(x)\big)$ is the anti-derivative (indefinite integral) of $f(x)$, that is

$$\frac{1}{D}\big(f(x)\big) = \int f(x)\,dx.$$

We apply the above formula to the next example.

Example 2.2.3

Solve $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} = 8$.

Solution

Find y_c: Consider $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} = 0$.

Characteristic equation: $m^2 - 4m = 0 \Rightarrow m = 0$ or $m = 4$.

$y_c = C_1e^0 + C_2e^{4x}$ or $y_c = C_1 + C_2e^{4x}$

Find y_p: Consider $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} = 8$.

$$(D^2 - 4D)y = 8 \Rightarrow y_p = \frac{1}{D^2 - 4D}(8)$$

$$\Rightarrow y_p = \frac{1}{D}\left(\frac{1}{D-4}(8)\right) = \frac{1}{D}\left(\frac{8}{0-4}\right) = \int -2\,dx = -2x$$

Thus, the general solution is

$$y = y_c + y_p = C_1 + C_2e^{4x} - 2x.$$

Inverse D-operator of an Exponential Function

$$\frac{1}{F(D)}(e^{kx}) = \frac{1}{F(k)}(e^{kx})$$

In the formula above, $F(k)$ is the value obtained by replacing D in $F(D)$ with k, provided that $F(k) \neq 0$. If $F(k) = 0$, we use a different formula (see Example 2.2.5).

Example 2.2.4

Solve $\dfrac{d^2y}{dx^2} - 4y = e^{4x}$.

Solution

Find y_c: Consider $\dfrac{d^2y}{dx^2} - 4y = 0$. From Example 2.1.2, we have

$$y_c = C_1e^{2x} + C_2e^{-2x}.$$

Find y_p: Consider $\dfrac{d^2y}{dx^2} - 4y = e^{4x}$.

$$(D^2 - 4)y = e^{4x} \Rightarrow y_p = \frac{1}{D^2 - 4}(e^{4x})$$

$$\Rightarrow y_p = \frac{1}{4^2 - 4}(e^{4x}) = \frac{1}{12}e^{4x}$$

Thus, the general solution is

$$y = y_c + y_p = C_1e^{2x} + C_2e^{-2x} + \frac{1}{12}e^{4x}.$$

Inverse D-operator of a Product involving Exponential Function

$$\frac{1}{F(D)}\left(e^{kx}\,V(x)\right) = e^{kx}\frac{1}{F(D+k)}\left(V(x)\right)$$

In the formula above, $F(D+k)$ is obtained by replacing D in $F(D)$ with $D+k$. This new inverse D-operator then acts on the input $V(x)$, which may be a constant, sine, or cosine function.

Example 2.2.5

Solve $\dfrac{d^2y}{dx^2} - 4y = e^{2x}$.

Solution

Find y_c: Consider $\dfrac{d^2y}{dx^2} - 4y = 0$. From Example 2.1.2, we have

$$y_c = C_1e^{2x} + C_2e^{-2x}.$$

Find y_p: Consider $\dfrac{d^2y}{dx^2} - 4y = e^{2x}$.

$$(D^2-4)y = e^{2x} \Rightarrow y_p = \frac{1}{D^2-4}(e^{2x})$$

$$\Rightarrow y_p = e^{2x}\frac{1}{(D+2)^2-4}(1) = e^{2x}\frac{1}{D^2+4D}(1)$$

$$\Rightarrow y_p = e^{2x}\frac{1}{D}\left(\frac{1}{D+4}(1)\right) = e^{2x}\int\frac{1}{4}dx = \frac{1}{4}x\,e^{2x}$$

Thus, the general solution is

$$y = y_c + y_p = C_1e^{2x} + C_2e^{-2x} + \frac{1}{4}xe^{2x}.$$

Inverse D-operators of Sine and Cosine Functions

$$\frac{1}{F(D^2)}(\sin kx) = \frac{1}{F(-k^2)}(\sin kx)$$

$$\frac{1}{F(D^2)}(\cos kx) = \frac{1}{F(-k^2)}(\cos kx)$$

In the formula above, obtaining $F(-k^2)$ from $F(D^2)$ means replacing only D^2 by $-k^2$, provided that $F(-k^2) \neq 0$. If $F(-k^2) = 0$, we use different formula (see Example 2.2.7).

Example 2.2.6

Solve $\dfrac{d^2y}{dx^2} + 4y = \sin x$.

Solution

Find y_c: Consider $\dfrac{d^2y}{dx^2} + 4y = 0$. From Example 2.1.4, we have

$$y_c = C_1 \sin 2x + C_2 \cos 2x.$$

Find y_p: Consider $\dfrac{d^2y}{dx^2} + 4y = \sin x$.

$$(D^2 + 4)y = \sin x \Rightarrow y_p = \frac{1}{D^2 + 4}(\sin x)$$

$$\Rightarrow y_p = \frac{1}{-1^2 + 4}(\sin x) = \frac{1}{3}\sin x$$

Thus, the general solution is

$$y = y_c + y_p = C_1 \sin 2x + C_2 \cos 2x + \frac{1}{3}\sin x.$$

Take note of the following special forms for the inverse D-operators of sine and cosine functions.

Special Forms for Inverse D-operators of Sine and Cosine Functions

$$\frac{1}{D^2+k^2}(\sin kx) = \frac{-x\cos kx}{2k}$$

$$\frac{1}{D^2+k^2}(\cos kx) = \frac{x\sin kx}{2k}$$

Example 2.2.7

Solve $\dfrac{d^2y}{dx^2} + 4y = \sin 2x$.

Solution

Find y_c: Consider $\dfrac{d^2y}{dx^2} + 4y = 0$. From Example 2.1.4, we have

$$y_c = C_1 \sin 2x + C_2 \cos 2x.$$

Find y_p: Consider $\dfrac{d^2y}{dx^2} + 4y = \sin 2x$.

$$(D^2+4)y = \sin 2x \Rightarrow y_p = \frac{1}{D^2+4}(\sin 2x)$$

$$\Rightarrow y_p = \frac{-x\cos 2x}{4}$$

Thus, the general solution is

$$y = y_c + y_p = C_1 \sin 2x + C_2 \cos 2x - \frac{x\cos 2x}{4}.$$

In the example below, the inverse D-operator contains both D^2 and D terms.

Example 2.2.8

Solve $\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} + 4y = \sin 2x$.

Solution

Find y_c: Consider $\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} + 4y = 0$. From Example 2.1.3, we have

$y_c = (C_1 x + C_2)\, e^{-2x}$.

Find y_p: Consider $\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} + 4y = \sin 2x$.

$$(D^2 + 4D + 4)y = \sin 2x \Rightarrow y_p = \frac{1}{D^2 + 4D + 4}(\sin 2x)$$

$$\Rightarrow y_p = \frac{1}{-2^2 + 4D + 4}(\sin 2x)$$

$$\Rightarrow y_p = \frac{1}{4D}(\sin 2x)$$

$$\Rightarrow y_p = \frac{1}{4}\int \sin 2x \ dx = -\frac{\cos 2x}{8}$$

Thus, the general solution is

$$y = y_c + y_p = (C_1 x + C_2)\, e^{-2x} - \frac{\cos 2x}{8}.$$

When dealing with sine or cosine functions, we consider only D^2 terms in the expression of the inverse D-operator. If the expression does not contain any D^2 terms, a simple modification using the *conjugate pair* is necessary.

Observe that $\dfrac{1}{D-n}\left[\dfrac{1}{D+n}(\sin kx)\right] = \dfrac{1}{D^2-n^2}(\sin kx)$ implies

$$\frac{1}{D+n}(\sin kx) = (D-n)\left[\frac{1}{D^2-n^2}(\sin kx)\right].$$

The roles of $(D+n)$ and $(D-n)$ are interchangeable.

Example 2.2.9

Solve $y'' + 4y' + 4y = \sin x$.

Solution

Find y_c: Consider $y'' + 4y' + 4y = 0$. From Example 2.1.3, we have

$y_c = (C_1 x + C_2)\, e^{-2x}$.

Find y_p: Consider $y'' + 4y' + 4y = \sin x$.

$$(D^2 + 4D + 4)y = \sin x \Rightarrow y_p = \frac{1}{D^2+4D+4}(\sin x)$$

$$\Rightarrow y_p = \frac{1}{-1^2+4D+4}(\sin x) = \frac{1}{4D+3}(\sin x)$$

$$\Rightarrow y_p = (4D-3)\left[\frac{1}{16D^2-9}(\sin x)\right] = (4D-3)\left[\frac{1}{-16-9}(\sin x)\right]$$

$$\Rightarrow y_p = \frac{4}{-25}D(\sin x) - \frac{3}{-25}(\sin x) = -\frac{4}{25}\cos x + \frac{3}{25}\sin x$$

Thus, the general solution is

$$y = y_c + y_p = (C_1 x + C_2)\, e^{-2x} - \frac{4}{25}\cos x + \frac{3}{25}\sin x.$$

The following example uses a combination of a few formulas.

Example 2.2.10
Solve $y'' + 4y' + 4y = e^{-2x}\sin x$.

Solution

Find y_c: Consider $y'' + 4y' + 4y = 0$. From Example 2.1.3, we have

$$y_c = (C_1 x + C_2)\, e^{-2x}.$$

Find y_p: Consider $y'' + 4y' + 4y = e^{-2x}\sin x$.

$$(D^2 + 4D + 4)y = e^{-2x}\sin x \Rightarrow y_p = \frac{1}{D^2 + 4D + 4}(e^{-2x}\sin x)$$

$$\Rightarrow y_p = e^{-2x}\frac{1}{(D-2)^2 + 4(D-2) + 4}(\sin x)$$

$$\Rightarrow y_p = e^{-2x}\frac{1}{(D^2 - 4D + 4) + (4D - 8) + 4}(\sin x)$$

$$\Rightarrow y_p = e^{-2x}\frac{1}{D^2}(\sin x)$$

$$\Rightarrow y_p = e^{-2x}\left(\frac{1}{-1}\sin x\right) = -e^{-2x}\sin x$$

Thus, the general solution is

$$y = y_c + y_p = (C_1 x + C_2)\, e^{-2x} - e^{-2x}\sin x.$$

2.3 Applications

In this section, we explore engineering problems modelled by second order linear differential equations. Examples in electronics and mechanics are presented together with brief descriptions on the modelling process.

Applications in Electronics

Recall the **Kirchoff's Voltage Law (KVL)** used in circuit analysis:

$$V_R + V_L + V_C = V_{\text{source}}.$$

Using the following formulas

$$V_R = iR, \qquad V_L = L\frac{di}{dt}, \qquad V_C = \frac{q}{C}, \qquad i = \frac{dq}{dt},$$

the KVL translates to the second order linear differential equation

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = V_{\text{source}}.$$

The next example is an electronics problem modelled by a linear differential equation involving $\dfrac{d^2q}{dt^2}$. We solve for the particular solution by considering initial charge $q(0)$ and initial current $i(0) = q'(0)$.

Example 2.3.1

A series circuit consists of a 1 H inductor, a 1 kΩ resistor, and a 6.25μF capacitor. The capacitor was initially charged at 1.5 mC. The circuit is then closed at time $t = 0$, allowing the capacitor to discharge, and satisfies the differential equation

$$\frac{d^2q}{dt^2} + 1000\frac{dq}{dt} + 160000q = 0.$$

Express the charge q in terms of time t.

Solution

Characteristic equation: $m^2 + 1000m + 160000 = 0 \Rightarrow m = -800$ or $m = -200$.

So, $q = C_1 e^{-800t} + C_2 e^{-200t}$.

$$q(0) = 0.0015 \Rightarrow 0.0015 = C_1 e^0 + C_2 e^0 \Rightarrow C_1 + C_2 = 0.0015$$

$$q' = -800\, C_1 e^{-800t} - 200\, C_2 e^{-200t}$$

$$q'(0) = 0 \Rightarrow 0 = -800\, C_1 e^0 - 200\, C_2 e^0 \Rightarrow 4C_1 + C_2 = 0$$

Solving the linear system for C_1 and C_2, we have $C_1 = -0.0005$ and $C_2 = 0.002$.

Thus, $q = -0.0005e^{-800t} + 0.002e^{-200t}$.

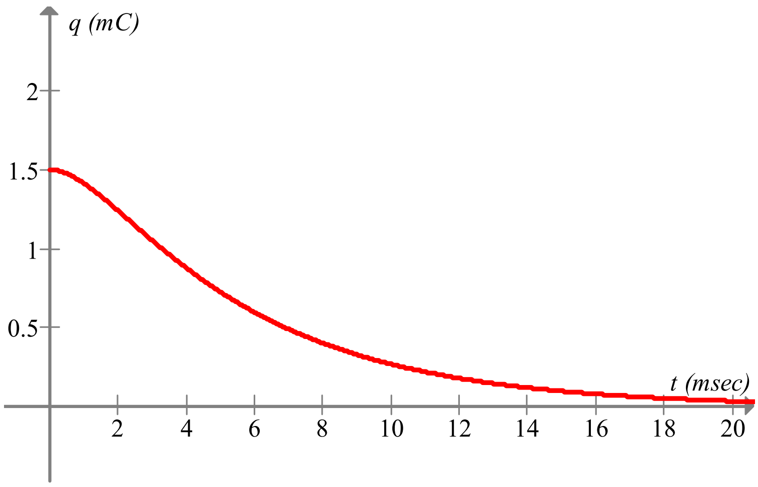
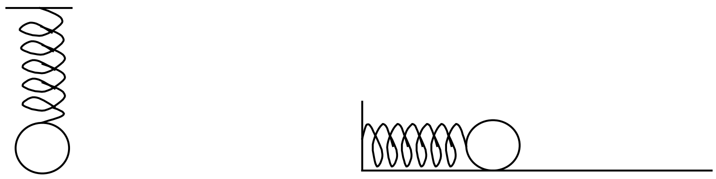


Figure 2.1. Graph of q (millicoulomb) vs. t (millisecond) in Example 2.3.1

Applications in Mechanics

An interesting topic in classical mechanics is the harmonic motion, e.g. an object moves back and forth due to the potential energy of a spring.



(a) vertical (b) horizontal

Figure 2.2 Vertical and horizontal harmonic montions

In studying harmonic motion, the **Hooke's Law**, $F_{spring} = -kx$, is considered. Here, x denotes the displacement of the object from its initial position, and k is a spring constant that determines the stiffness of the spring.

In the next two examples, we see problems in harmonic motion modelled by second order differential equations involving $\dfrac{d^2x}{dt^2}$. We solve for the particular solution by considering initial displacement $x(0)$ and initial velocity $v(0) = x'(0)$.

Example 2.3.2
A spring with spring constant of 1 N/m is lying horizontally on a smooth surface with one end fixed to a wall as shown in Figure 2.2(b) above. An object of unit mass is attached to the other end of the spring without stretching it. Let x be the displacement of the object from its equilibrium position. The object projected horizontally away from the wall with initial velocity 1 m/s satisfies the differential equation

$$\frac{d^2x}{dt^2} + x = 0.$$

Express the displacement x as a function of time t.

Solution

Characteristic equation: $m^2 + 1 = 0 \Rightarrow m = j$ or $m = -j$.

So, $x = C_1 \sin t + C_2 \cos t$.

Initially, the spring is not stretched $\Rightarrow x(0) = 0$.

$x(0) = 0 \Rightarrow 0 = C_1 \sin 0 + C_2 \cos 0 \Rightarrow C_2 = 0$

Initial velocity is 1 m/s $\Rightarrow x'(0) = 1$.

$x' = C_1 \cos t - C_2 \sin t$

$x'(0) = 1 \Rightarrow 1 = C_1 \cos 0 - C_2 \sin 0 \Rightarrow C_1 = 1$

Thus, $x = \sin t$.

The solution in Example 2.3.2 indicates that the object is moving back and forth, pushing and pulling the spring alternately forever. This is an ideal situation with no resistance involved in the harmonic motion. In a real situation, this is not possible.

Example 2.3.3

A system consists of a spring of stiffness 0.625 N/m and a 25 gram ball suspended by the spring as shown in Figure 2.2(a) in the previous page. The other end of the spring is fixed onto a ceiling. The ball is pulled downwards 50 cm from its equilibrium position, and released. Let v be the velocity of the ball, and x its displacement from its equilibrium position. The system experiences a resistance $0.25v$, and satisfies the differential equation

$$\frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 25x = 0.$$

Express the displacement x as a function of time t.

Solution

Characteristic equation: $m^2 + 10m + 25 = 0 \Rightarrow m = -5.$

So, $x = (C_1 + C_2\, t)e^{-5t}$.

Initially, the spring is pulled 50 cm downwards $\Rightarrow x(0) = -0.5.$

$x(0) = -0.5 \Rightarrow -0.5 = (C_1 + 0)e^0 \Rightarrow C_1 = -0.5$

The ball was released after being pulled $\Rightarrow x'(0) = 0.$

$x' = (-5\, C_1 - 5\, C_2 t + C_2)e^{-5t}$

$x'(0) = 0 \Rightarrow 0 = (-5\, C_1 - 0 + C_2)e^0 \Rightarrow C_2 = -2.5$

Thus, $x = (-0.5 - 2.5\, t)e^{-5t}$.

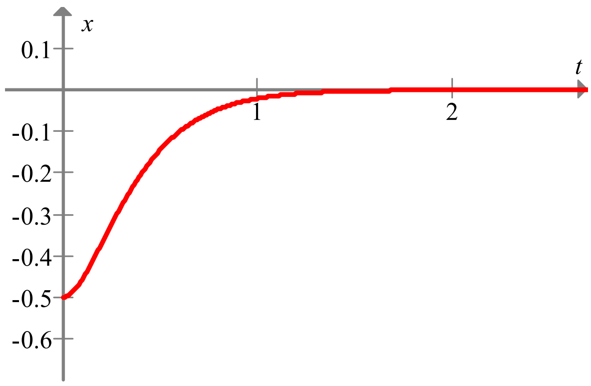


Figure 2.2. Graph of x vs. t for harmonic motion in Example 2.3.3

2.4 Exercises

Part A: Baseline Problems

1. Solve the following homogeneous second order linear ODEs, see Section 2.1.

   (a) $y'' + 4y' - 5y = 0$

   (b) $y'' + 6y' + 9y = 0$

   (c) $y'' - 16y = 0$

   (d) $\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} + 2y = 0$

   (e) $6\dfrac{d^2y}{dx^2} - 11\dfrac{dy}{dx} + 3y = 0$

2. Solve the following homogeneous second order linear ODEs, see Section 2.1.

   (a) $\dfrac{d^2x}{dt^2} + \dfrac{dx}{dt} = 0$

   (b) $\dfrac{d^2x}{dt^2} - 6\dfrac{dx}{dt} + 10x = 0$

   (c) $\dfrac{d^2q}{dt^2} + 4q = 0$

   (d) $\dfrac{d^2q}{dt^2} + 2\dfrac{dq}{dt} - q = 0$

   (e) $4\dfrac{d^2z}{dx^2} - 12\dfrac{dz}{dx} + 9z = 0$

Part B: Intermediate Problems

3. Solve the following Initial Value Problems (IVP), see Example 2.1.5.

(a) $$\begin{cases} y'' + 4y' - 5y = 0 \\ y(0) = 1 \\ y'(0) = 1 \end{cases}$$

(b) $$\begin{cases} \dfrac{d^2x}{dt^2} + \dfrac{dx}{dt} = 0 \\ x(0) = 0 \\ x'(0) = 1 \end{cases}$$

(c) $$\begin{cases} \dfrac{d^2q}{dt^2} + 4q = 0 \\ q(0) = 1 \\ q'(0) = 0 \end{cases}$$

For Questions 4 – 6, see Examples 2.2.2 and 2.2.3.

4. (a) Find $\dfrac{1}{D^2 + 4D - 5}(15)$.

(b) Solve $y'' + 4y' - 5y = 15$.

5. (a) Find $\dfrac{1}{D^2 + 6D + 9}(21)$.

(b) Solve $y'' + 6y' + 9y = 21$.

6. (a) Find $\dfrac{1}{D^2+D}(2)$.

(b) Solve $\dfrac{d^2x}{dt^2}+\dfrac{dx}{dt}=2$.

For Questions 7 – 9, see Example 2.2.4.

7. (a) Find $\dfrac{1}{D^2+6D+9}(e^x)$.

(b) Solve $y''+6y'+9y=8e^x$.

8. (a) Find $\dfrac{1}{D^2-1}(e^{2x})$.

(b) Solve $y''-y=e^{2x}$.

9. (a) Find $\dfrac{1}{D^2+D}(e^{5t})$.

(b) Solve $\dfrac{d^2x}{dt^2}+\dfrac{dx}{dt}=20e^{5t}$.

For Questions 10 – 11, see Example 2.2.5.

10. (a) Find $\dfrac{1}{D^2+6D+9}(e^{-3x})$.

(b) Solve $y''+6y'+9y=e^{-3x}$.

11. (a) Find $\dfrac{1}{D^2 + D}(e^{-t})$.

    (b) Solve $\dfrac{d^2x}{dt^2} + \dfrac{dx}{dt} = e^{-t}$.

For Questions 12 – 14, see Examples 2.2.6 and 2.2.7.

12. (a) Find $\dfrac{1}{D^2 - 1}(4\cos x)$.

    (b) Solve $y'' - y = 4\cos x$.

13. (a) Find $\dfrac{1}{D^2 + 1}(4\cos x)$.

    (b) Solve $y'' + y = 4\cos x$.

14. (a) Find $\dfrac{1}{D^2 + 9}(\cos 3t)$.

    (b) Solve $\dfrac{d^2q}{dt^2} + 9q = \cos 3t$

For Questions 15 – 17, see Examples 2.2.8 and 2.2.9.

15. (a) Find $\dfrac{1}{D^2 + 6D + 9}(\sin 3x)$.

    (b) Solve $y'' + 6y' + 9y = \sin 3x$.

16. (a) Find $\dfrac{1}{D^2+2D+2}(\sin x)$.

(b) Solve $y''+2y'+2y=\sin x$.

17. (a) Find $\dfrac{1}{D^2+D}(\cos 3t)$.

(b) Solve $\dfrac{d^2x}{dt^2}+\dfrac{dx}{dt}=\cos 3t$.

For Questions 18 – 19, see Example 2.2.10.

18. (a) Find $\dfrac{1}{D^2+6D+9}(e^{-3x}\sin x)$.

(b) Solve $y''+6y'+9y=e^{-3x}\sin x$.

19. (a) Find $\dfrac{1}{D^2+2D+2}(e^{-x}\cos x)$.

(b) Solve $y''+2y'+2y=e^{-x}\cos x$.

20. Solve the following non-homogeneous second order linear ODEs.

(a) $y''+4y=\sin x+e^{2x}+1$

(b) $y''-16y=6\sin 2x-8\cos x+4$

(c) $y''+6y'+9y=\sin 3x-2e^{-3x}$

Part C: Application Problems

See Section 2.3.

21. A series circuit consists of a capacitor of capacitance 0.4 mF, an inductor of inductance 1 H, a resistor of resistance 0.1 kΩ. The capacitor was initially charged at 0.01 C. The circuit is then closed at time $t = 0$ and satisfies the differential equation

$$\frac{d^2q}{dt^2} + 100\frac{dq}{dt} + 2500q = 0.$$

Express the charge q in terms of t.

22. A 2 kg metal ball is suspended by a spring of stiffness 10 N/m. The ball is pulled down 0.1 m below its equilibrium position and released. Let x be the displacement of the object from its equilibrium position. The harmonic motion of the ball is subjected to air resistance, and satisfies the differential equation

$$\frac{d^2x}{dt} + 2\frac{dx}{dt} + 5x = 0.$$

Express the displacement x as a function of t.

– END OF CHAPTER 2 –